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T11 - 05

Assignment No. 7

Aim: Implementation of Bayesian Classifier

Program:

```
import pandas as pd

class BayesianClassifier:
    def __init__(self):
        self.prob_yes = 0 #prior probability
        self.prob_no = 0  #prior probability
        self.conditional_prob = {}

    def train(self, data):
        # Split data into attributes and labels
        attributes = data.iloc[:, :-1].values
        labels = data.iloc[:, -1].values

        total_yes = sum(1 for label in labels if label == 'yes')
        total_no = len(labels) - total_yes
        total = len(labels)

        self.prob_yes = total_yes / total # prior probability of the class "yes"
(i.e., the proportion of instances labeled "yes").
        self.prob_no = total_no / total #prior probability of the class "no"

        # Calculate conditional probabilities
        num_attributes = attributes.shape[1] #number of features (attributes)

        for i in range(num_attributes):
            self.conditional_prob[i] = {}

            for value in set(attributes[:, i]): # unique values of the feature i
in the dataset.
                # The number of occurrences where feature i has a specific value,
and the class label is "yes".
                count_yes = sum(1 for j in range(total) if attributes[j, i] ==
value and labels[j] == 'yes')
                count_no = sum(1 for j in range(total) if attributes[j, i] ==
value and labels[j] == 'no')
```

```python
                #For each feature value, the conditional probability of the
feature value given the class label ("yes" or "no") is stored
                self.conditional_prob[i][value] = {
                    'yes': count_yes / total_yes if total_yes > 0 else 0,
#probability of the feature value occurring given the class "yes".
                    'no': count_no / total_no if total_no > 0 else 0
                }
    # predict method takes an instance (a list of feature values) and calculates
the posterior probability for each class ("yes" and "no")
    def predict(self, instance):
        #prior probability of the class "yes", which is multiplied by the
conditional probabilities of the instance’s feature values given the class "yes".
        prob_instance_yes = self.prob_yes
        prob_instance_no = self.prob_no

        for i, value in enumerate(instance):
            if value in self.conditional_prob[i]:
                prob_instance_yes *= self.conditional_prob[i][value]['yes']
                prob_instance_no *= self.conditional_prob[i][value]['no']

        return 'yes' if prob_instance_yes > prob_instance_no else 'no'


dataset = pd.read_csv('Buy_Computer.csv')

classifier = BayesianClassifier()
classifier.train(dataset)

test_instance = ['middle_age', 'high', 'no', 'fair']
prediction = classifier.predict(test_instance)

print("Predicted class:", prediction)
```

Output:

```
[6]   ✓ 0.0s

...   Predicted class: yes
```